

Quick Check

Lesson 5: Inside the Fruit: Placentation as Evidence of Post-Fertilisation Development

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 2.1: Reproduction in Plants | Lesson 5

After a flower has been fertilised, which part of the flower becomes the fruit and which part becomes the seed?

- A** The ovule becomes the fruit and the ovary becomes the seed
- B** The ovary becomes the fruit and the ovule becomes the seed
- C** The ovary becomes the seed and the stigma becomes the fruit
- D** The petals become the fruit and the ovule becomes the seed

What is placentation in a fruit?

- A** The way a fruit changes colour and softens as it ripens
- B** The way pollen lands on the stigma of a flower
- C** The number of seeds that a fruit can produce
- D** The way ovules or seeds are attached to the placenta inside the ovary or fruit

A learner cuts a tomato across (transverse section). She sees 2 to 3 chambers, and the seeds are attached to a central column where the walls between the chambers meet. Which type of placentation does the tomato show?

- A** Axile
- B** Free-central
- C** Parietal
- D** Marginal

A passion fruit is cut across. It has one single chamber and the seeds are attached to the inner wall of the fruit. Which type of placentation is this?

- A** Axile, because the seeds are attached to a central axis
- B** Marginal, because the seeds are attached along one seam
- C** Free-central, because there is a column in the middle
- D** Parietal, because the seeds are attached to the wall of a single chamber

A lone pawpaw tree in a compound is covered in flowers but bears no fruit, and there is no other pawpaw tree nearby. Which explanation is best?

- A** The ovules turned into fruit without the ovary, so no seeds could form
- B** The petals normally become the fruit, and they fell off the tree too early
- C** The flowers were not pollinated and fertilised, so the ovules did not become seeds and the ovary did not become a fruit
- D** The tree has the wrong placentation, so the flowers cannot produce fruit

A learner says, "A bean pod is not a fruit because it is dry and not sweet like a mango." Which response is correct?

- A** The learner is wrong, because a bean pod is a seed that has grown larger
- B** The learner is wrong, because the bean pod is a dry fruit: it is the developed ovary and it contains the seeds
- C** The learner is right, because the pod is a leaf that protects the seeds
- D** The learner is right, because only sweet and juicy structures are fruits

A farmer plants a piece of cassava stem and it grows into a new plant. Which statement correctly describes how this new plant formed?

- A** It formed after the cutting was pollinated by pollen from another cassava plant
- B** It formed because the cutting first made seeds inside a fruit
- C** It formed without flowers, fertilisation or seeds, because the cutting grows directly into a new plant
- D** It formed after double fertilisation inside the cutting

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).